

Unit Test 1: Relations and Functions

Class: 10

Total Marks: 50

Subject: Mathematics

Time: 1.30 hours

Part - A)

 $(8 \times 1 = 8)$

Answer all the questions.

- If $n(A \times B) = 6$ and $A = \{1, 3\}$, then $n(B)$ is
(a) 1 (b) 2 (c) 3 (d) 6
- If $A = \{a, b, p\}$, $B = \{2, 3\}$, $C = \{p, q, r, s\}$ then $n[(A \cup C) \times B]$ is
(a) 8 (b) 20 (c) 12 (d) 16
- If $A = \{1, 2\}$, $B = \{1, 2, 3, 4\}$, $C = \{5, 6\}$ and $D = \{5, 6, 7, 8\}$ then which of the following is true?
(a) $(A \times C) \subset (B \times D)$ (b) $(B \times D) \subset (A \times C)$
(c) $(A \times B) \subset (A \times D)$ (d) $(D \times A) \subset (B \times A)$
- The range of the relation $R = \{(x, x^2) \mid x \text{ is a prime number less than } 13\}$ is
(a) $\{2, 3, 5, 7\}$ (b) $\{2, 3, 5, 7, 11\}$
(c) $\{4, 9, 25, 49, 121\}$ (d) $\{1, 4, 9, 25, 49, 121\}$
- If the ordered pairs $(a + 2, 4)$ and $(5, 2a + b)$ are equal, then (a, b) is
(a) $(2, -2)$ (b) $(5, 1)$ (c) $(2, 3)$ (d) $(3, -2)$
- If $f(x) = 2x^2$ and $g(x) = \frac{1}{3x}$, then $f \circ g$ is
(a) $\frac{3}{2x^2}$ (b) $\frac{2}{3x^2}$ (c) $\frac{2}{9x^2}$ (d) $\frac{1}{6x^2}$
- If $f(x) = x + 1$ and $g(x) = x - 1$, find $f \circ g(x)$.
(a) $x + 1$ (b) $x - 1$ (c) x (d) x^2
- If $A = \{1, 2, 3\}$ and $B = \{a, b, c\}$, then the number of one-to-one functions from A to B is:
(a) 3 (b) 6 (c) 9 (d) 27

(Part - B)

 $(6 \times 2 = 12)$

Answer any 6 questions. (Question No. 16 is compulsory).

- If $A = \{1, 3, 5\}$ and $B = \{2, 3\}$, find $A \times B$ and $B \times A$.
- If $A \times B = \{(3, 2), (3, 4), (5, 2), (5, 4)\}$, find A and B .
- Let $X = \{1, 2, 3, 4\}$, $Y = \{2, 4, 6, 8, 10\}$ and $R = \{(1, 2), (2, 4), (3, 6), (4, 8)\}$. Show that R is a function. Also find its range.
- If $f(x) = 2x - x^2$, find $f(1)$ and $f(x + 1)$.

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13. If $f(x) = 3x - 2$, $g(x) = 2x + k$ and $f \circ g = g \circ f$, then find the value of k .
14. If $f(x) = x - 6$ and $g(x) = x^2$, find $f \circ g$ and $g \circ f$.
15. Represent the function $f = \{(1, 2), (2, 2), (3, 2), (4, 3), (5, 4)\}$ through a table.
16. **(Compulsory Question)** Let $A = \{x \in \mathbb{N} \mid x < 5\}$ and $B = \{x \in \mathbb{W} \mid x \text{ is an even number, } x \leq 6\}$. Verify that $n(A \times B) = n(A) \times n(B)$.

(Part - C)

(6 × 5 = 30)

Answer any 6 questions. (Question No. 24 is compulsory).

17. Let $A = \{x \in \mathbb{N} \mid 1 < x < 4\}$, $B = \{x \in \mathbb{W} \mid 0 \leq x < 2\}$ and $C = \{x \in \mathbb{N} \mid x < 3\}$. Verify that $A \times (B \cup C) = (A \times B) \cup (A \times C)$.
18. Let $A = \{1, 2, 3, 4\}$ and $B = \{2, 5, 8, 11, 14\}$ be two sets. A function $f : A \rightarrow B$ is given by $f(x) = 3x - 1$. Represent this function using (i) an arrow diagram (ii) a table (iii) a set of ordered pairs (iv) a graphical form.
19. If $f(x) = 2x + 3$, $g(x) = 1 - 2x$ and $h(x) = 3x$, prove that $f \circ (g \circ h) = (f \circ g) \circ h$.
20. Let A be the set of all natural numbers less than 8, B be the set of all prime numbers less than 8 and C be the set of even prime numbers. Verify that $(A \cap B) \times C = (A \times C) \cap (B \times C)$.
21. If a function $f : \mathbb{R} \rightarrow \mathbb{R}$ is defined by

$$f(x) = \begin{cases} 2x + 7; & x < -2 \\ x^2 - 2; & -2 \leq x < 3 \\ 3x - 2; & x \geq 3 \end{cases}$$

then find the values of (i) $f(4)$ (ii) $f(-2)$ (iii) $f(4) + 2f(1)$ (iv) $\frac{f(1)-3f(4)}{f(-3)}$.

22. If $f(x) = x^2$, $g(x) = 3x$ and $h(x) = x - 2$, prove that $(f \circ g) \circ h = f \circ (g \circ h)$.
23. Forensic scientists can determine a person's height (in cm) from the length of their thigh bone. They usually use the function $h(b) = 2.47b + 54.10$, where b is the length of the thigh bone. (i) Verify if h is a one-to-one function. (ii) If the length of the thigh bone is 50 cm, find the height of the person. (iii) If the height of the person is 147.96 cm, find the length of their thigh bone.
24. **(Compulsory Question)** For the linear functions $f(x) = ax + b$ and $g(x) = cx + d$, if $f \circ g = g \circ f$, prove that $f(d) = g(b)$. Also, if $f(x) = 3x + 2$ and $g(x) = cx - 1$ such that $f \circ g = g \circ f$, find the value of c .

Unit Test 2: Numbers and Sequences

Class: 10

Total Marks: 50

Subject: Mathematics

Time: 1.30 hours

(Part - A) $(8 \times 1 = 8)$

Answer all the questions.

- According to Euclid's division lemma, for any positive integers a and b , there exist unique integers q and r such that $a = bq + r$, where r must satisfy:
 (a) $1 < r < b$ (b) $0 < r < b$ (c) $0 \leq r < b$ (d) $0 < r \leq b$
- When finding the HCF of 65 and 117 expressed in the form $65m - 117$, the value of m is:
 (a) 4 (b) 2 (c) 1 (d) 3
- If 6 times the 6th term of an A.P. is equal to 7 times the 7th term, then the 13th term of the A.P. is:
 (a) 0 (b) 6 (c) 7 (d) 13
- The next term of the sequence $\frac{3}{16}, \frac{1}{8}, \frac{1}{12}, \frac{1}{18}$ is:
 (a) $\frac{1}{24}$ (b) $\frac{1}{27}$ (c) $\frac{2}{3}$ (d) $\frac{1}{81}$
- If $t_1, t_2, t_3, \dots$ are in an A.P., then $t_6, t_{12}, t_{18}, \dots$ form:
 (a) a Geometric Progression (b) an Arithmetic Progression
 (c) Neither (d) a Constant sequence
- The value of $(1^3 + 2^3 + 3^3 + \dots + 15^3) - (1 + 2 + 3 + \dots + 15)$ is:
 (a) 14400 (b) 14200 (c) 14280 (d) 14520
- If n is any positive integer, then $n^2 + n$ is always:
 (a) an odd number (b) an even number (c) a multiple of 3 (d) a multiple of 5
- If a, b, c are in G.P. and $a > 0, b > 0, c > 0$, then $\log a, \log b, \log c$ are in:
 (a) Arithmetic Progression (b) Geometric Progression
 (c) Both (d) None

(Part - B) $(6 \times 2 = 12)$

Answer any 6 questions. (Question No. 16 is compulsory).

- Use Euclid's Division Algorithm to find the Highest Common Factor (HCF) of 340 and 412.
- What is the time 100 hours after 7 a.m.?
- Find the next three terms of the sequence $\frac{1}{4}, \frac{2}{9}, \frac{3}{16}, \dots$
- Find the middle terms of the A.P. 9, 15, 21, 27, ..., 183.

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13. Find the sum to infinity of the series $3 + 1 + \frac{1}{3} + \dots \infty$.
14. If $1 + 2 + 3 + \dots + k = 325$, find $1^3 + 2^3 + 3^3 + \dots + k^3$.
15. Compute the value of x such that $10^4 \equiv x \pmod{19}$.
16. **(Compulsory Question)** A bouncing ball is dropped from a height of 48 meters. After each bounce, it rebounds to two-thirds of the height from which it fell. Find the total distance travelled by the ball before it comes to rest.

((Part - C)

(6 × 5 = 30)

Answer any 6 questions. (Question No. 24 is compulsory).

17. Find the HCF of 396, 504, 636 using Euclid's division algorithm.
18. If 9 times the 9th term is equal to 15 times the 15th term of an A.P., show that 6 times the 24th term is zero.
19. The houses of a street are numbered from 1 to 49. Senthil's house is numbered such that the sum of numbers of the houses prior to Senthil's house is equal to the sum of numbers of the houses following Senthil's house. Find Senthil's house number.
20. The product of three consecutive terms of a Geometric Progression is 343 and their sum is $\frac{91}{3}$. Find the three terms.
21. Find the sum to n terms of the series $5 + 55 + 555 + \dots$
22. Rekha has 15 square colour papers of sizes 10 cm, 11 cm, 12 cm,..., 24 cm. How much area can be decorated with these colour papers?
23. If S_1, S_2 and S_3 are the sum of first $n, 2n$ and $3n$ terms of an Arithmetic Progression respectively, prove that $S_3 = 3(S_2 - S_1)$.
24. **(Compulsory Question)** If the sum to infinity of a G.P. is 15 and the sum of the squares of its terms is 45, find the G.P.

Unit Test 3: Algebra

(From Exercise 3.1 to 3.14)

Class: 10

Total Marks: 50

Subject: Mathematics

Time: 1.30 hours

(Part - A)

(8 × 1 = 8)

Answer all the questions.

- If a system of three linear equations in three variables has no solution, then the planes in the system:
 - intersect at a single point
 - intersect along a single line
 - coincide
 - do not intersect
- The solution of the system $x + y - 3z = -6$, $-7y + 7z = 7$, $3z = 9$ is:
 - $x = 1, y = 2, z = 3$
 - $x = -1, y = 2, z = 3$
 - $x = -1, y = -2, z = 3$
 - $x = 1, y = -2, z = 3$
- If the GCD of $x^2 - 2x - 24$ and $x^2 - kx - 6$ is $(x - 6)$, then the value of k is:
 - 3
 - 5
 - 6
 - 8
- $\frac{3y-3}{y} \div \frac{7y-7}{3y^2}$ is:
 - $\frac{9y}{7}$
 - $\frac{9y^3}{21y-21}$
 - $\frac{21y^2-42y+21}{3y^3}$
 - $\frac{7(y^2-2y+1)}{y^2}$
- The reduced form of $\frac{x}{x^2-25} - \frac{8}{x^2+6x+5}$ is:
 - $\frac{x^2-7x+40}{(x-5)(x+5)}$
 - $\frac{x^2+7x+40}{(x-5)(x+5)(x+1)}$
 - $\frac{x^2-7x+40}{(x^2-25)(x+1)}$
 - $\frac{x^2+10}{(x^2-25)(x+1)}$
- The square root of $\frac{256x^8y^4z^{10}}{25x^6y^6z^6}$ is:
 - $\frac{16}{5} \left| \frac{x^2z^4}{y^2} \right|$
 - $16 \left| \frac{y^2}{x^2z^4} \right|$
 - $\frac{16}{5} \left| \frac{y}{xz^2} \right|$
 - $\frac{16}{5} \left| \frac{xz^2}{y} \right|$
- What must be added to $x^4 + 64$ to make it a perfect square?
 - $4x^2$
 - $16x^2$
 - $8x^2$
 - $-8x^2$
- The solution of $(2x - 1)^2 = 9$ is:
 - 1
 - 2
 - 1, 2
 - None of these

(Part - B)

(6 × 2 = 12)

Answer any 6 questions. (Question No. 16 is compulsory).

- Solve the system of linear equations: $x + y + z = 5$; $2x - y + z = 9$; $x - 2y + 3z = 16$.
- Find the LCM of the polynomials $4x^2y, 8x^3y^2$.

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11. Simplify $\frac{p^2-10p+21}{p-7} \times \frac{p^2+p-12}{(p-3)^2}$.
12. Simplify $\frac{x^2-16}{x+4} \div \frac{x-4}{x+4}$.
13. Find the square root of the expression $4x^2 + 20x + 25$.
14. Solve the quadratic equation $x^2 - 3x - 2 = 0$ by completing the square method.
15. Solve the quadratic equation $2x^2 - 5x + 2 = 0$ using the formula method.
16. **(Compulsory Question)** If α and β are the roots of the equation $3x^2 + 7x - 2 = 0$, find the value of $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$.

Part - C)

(6 × 5 = 30)

Answer any 6 questions. (Question No. 24 is compulsory).

17. The average age of grandfather, father and Vani is 53. One-half of grandfather's age plus one-third of father's age plus one-fourth of Vani's age is 65. Four years ago, grandfather's age was four times Vani's age. Find their current ages.
18. Find the GCD of the polynomials $x^4 + 3x^3 - x - 3$ and $x^3 + x^2 - 5x + 3$.
19. If $A = \frac{2x+1}{2x-1}$ and $B = \frac{2x-1}{2x+1}$, find $\frac{1}{A-B} - \frac{2B}{A^2-B^2}$.
20. Arul, Madhan and Ram working together can clean a store in 6 hours. Working alone, Madhan takes twice as long to clean the store as Arul does. Ram needs three times as long as Arul does. How long would it take each of them to clean the store working alone?
21. Find the square root of $64x^4 - 16x^3 + 17x^2 - 2x + 1$ by division method.
22. If $9x^4 + 12x^3 + 28x^2 + ax + b$ is a perfect square, find the values of a and b .
23. A 17 ft long ladder is leaning against a wall. The ground, the ladder and the vertical wall form a right-angled triangle. If the distance from the foot of the wall to the foot of the ladder is 7 ft less than the height where the ladder touches the wall, find the height of the wall.
24. **(Compulsory Question)** Show that the equation $x^2(p^2 + q^2) + 2x(pr + qs) + r^2 + s^2 = 0$ has no real roots. Furthermore, if $ps = qr$, show that the roots are real and equal.

Unit Test 4: Geometry and Trigonometry

(Portions: Geometry 4.1, 4.2 and Trigonometry 6.1)

Class: 10

Total Marks: 50

Subject: Mathematics

Time: 1.30 hours

(Part - A)

(8 × 1 = 8)

Answer all the questions.

- When will the triangles ABC and EDF be similar if $\frac{AB}{DE} = \frac{BC}{FD}$?
 (a) $\angle B = \angle E$ (b) $\angle A = \angle D$ (c) $\angle B = \angle D$ (d) $\angle A = \angle F$
- In an isosceles triangle $\triangle ABC$, $\angle C = 90^\circ$ and $AC = 5$ cm, then AB is:
 (a) 2.5 cm (b) 5 cm (c) 10 cm (d) $5\sqrt{2}$ cm
- In $\triangle ABC$, $DE \parallel BC$. If $AB = 3.6$ cm, $AC = 2.4$ cm and $AD = 2.1$ cm, then the length of AE is:
 (a) 1.4 cm (b) 1.8 cm (c) 1.2 cm (d) 1.05 cm
- The value of $\sin^2 \theta + \frac{1}{1+\tan^2 \theta}$ is:
 (a) $\tan^2 \theta$ (b) 1 (c) $\cot^2 \theta$ (d) 0
- If $x = a \tan \theta$ and $y = b \sec \theta$ then:
 (a) $\frac{y^2}{b^2} - \frac{x^2}{a^2} = 1$ (b) $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ (c) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ (d) $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 0$
- The value of $(1 + \tan \theta + \sec \theta)(1 + \cot \theta - \csc \theta)$ is:
 (a) 2 (b) 1 (c) 0 (d) -1
- If the ratio of corresponding sides of two similar triangles is 4 : 9, then the ratio of their areas is:
 (a) 2 : 3 (b) 4 : 9 (c) 16 : 81 (d) 81 : 16
- Which of the following is equivalent to $\sec^2 \theta + \csc^2 \theta$?
 (a) $\sec^2 \theta \cdot \csc^2 \theta$ (b) $\sin^2 \theta \cdot \cos^2 \theta$ (c) 1 (d) $\tan^2 \theta - \cot^2 \theta$

(Part - B)

(6 × 5 = 30)

Answer any 6 questions. (Question No. 16 is compulsory).

- In $\triangle ABC$, points D and E are on the sides AB and AC respectively such that $DE \parallel BC$. If $\frac{AD}{DB} = \frac{3}{4}$ and $AC = 15$ cm, find the value of AE .
- The perimeters of two similar triangles $\triangle ABC$ and $\triangle PQR$ are 36 cm and 24 cm respectively. If $PQ = 10$ cm, find AB .
- In $\triangle ABC$, AD is the bisector of $\angle A$. If $BD = 4$ cm, $DC = 3$ cm and $AB = 6$ cm, find AC .
- Prove that $\tan^2 \theta - \sin^2 \theta = \tan^2 \theta \sin^2 \theta$.

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13. Prove that $\frac{\cos \theta}{1+\sin \theta} = \sec \theta - \tan \theta$.
14. Prove that $\frac{\sin A}{1+\cos A} + \frac{\sin A}{1-\cos A} = 2 \csc A$.
15. Prove that $\sec \theta - \cos \theta = \tan \theta \sin \theta$.
16. **(Compulsory Question)** If $\sec \theta - \tan \theta = \frac{1}{3}$, find the value of $\sec \theta + \tan \theta$. Hence compute the value of $\cos \theta$.

(Part - C)

(6 × 5 = 30)

Answer any 6 questions. (Question No. 24 is compulsory).

17. State and prove the Basic Proportionality Theorem (Thales Theorem).
18. State and prove the Angle Bisector Theorem.
19. Two poles of height a metres and b metres are p metres apart. Prove that the height of the point of intersection of the lines joining the top of each pole to the foot of the opposite pole is given by $\frac{ab}{a+b}$ metres.
20. In a trapezium $ABCD$, $AB \parallel DC$. E and F are points on non-parallel sides AD and BC respectively such that $EF \parallel AB$. Prove that $\frac{AE}{ED} = \frac{BF}{FC}$.
21. Prove that $\frac{\cos \theta}{1+\sin \theta} = \frac{1}{a}$, then $\frac{a^2-1}{a^2+1} = \sin \theta$.
22. If $\sec \theta + \tan \theta = p$, prove that $\sec \theta = \frac{p^2+1}{2p}$ and $\tan \theta = \frac{p^2-1}{2p}$. Hence show that $\sin \theta = \frac{p^2-1}{p^2+1}$.
23. Prove that $\frac{\tan^2 A - 1}{\tan^2 A + 1} = 1 - 2 \cos^2 A$.
24. **(Compulsory Question)** Prove the following trigonometric identity:

$$\frac{\sin^3 A + \cos^3 A}{\sin A + \cos A} + \frac{\sin^3 A - \cos^3 A}{\sin A - \cos A} = 2$$

Unit Test 5: Coordinate Geometry

Class: 10

Total Marks: 50

Subject: Mathematics

Time: 1.30 hours

(Part - A)**(8 × 1 = 8)***Answer all the questions.*

- The area of the triangle formed by the points $(-5,0)$, $(0,-5)$ and $(5,0)$ is:
 (a) 0 sq. units (b) 25 sq. units (c) 5 sq. units (d) None of these
- A person is walking near a wall such that the distance between him and the wall is 10 units. If the wall is considered as the Y-axis, then the path travelled by the person is:
 (a) $x = 10$ (b) $y = 10$ (c) $x = 0$ (d) $y = 0$
- The straight line given by the equation $x = 11$ is:
 (a) parallel to X-axis (b) parallel to Y-axis
 (c) passing through the origin (d) passing through the point $(0,11)$
- If $(5,7)$, $(3,p)$ and $(6,6)$ are collinear, then the value of p is:
 (a) 3 (b) 6 (c) 9 (d) 12
- The point of intersection of the straight lines $3x - y = 4$ and $x + y = 8$ is:
 (a) $(2,4)$ (b) $(5,3)$ (c) $(3,5)$ (d) $(4,4)$
- The slope of the line perpendicular to the line joining the points $(0,0)$ and $(-8,8)$ is:
 (a) -1 (b) $\sqrt{3}$ (c) 1 (d) 2
- The area of the triangle formed by the straight line $\frac{x}{a} + \frac{y}{b} = 1$ with the coordinate axes is:
 (a) ab sq. units (b) $\frac{1}{2}ab$ sq. units (c) $2ab$ sq. units (d) $a + b$ sq. units
- If the slope of a straight line is undefined, then the line is:
 (a) parallel to X-axis (b) parallel to Y-axis
 (c) passing through the origin (d) intersecting both coordinate axes

(Part - B)**(6 × 2 = 12)***Answer any 6 questions. (Question No. 16 is compulsory).*

- Find the area of the triangle formed by the points $(1,-1)$, $(-4,6)$ and $(-3,-5)$.
- If the angle of inclination of a line is 30° , find its slope.
- If the slope of the line joining $(-2,a)$ and $(9,3)$ is $-1/2$, find the value of a .
- Find the intercepts made by the line $3x - 2y - 6 = 0$ on the coordinate axes.

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13. Find the equation of a straight line passing through the point (5,7) and parallel to the X-axis.
14. Check whether the straight lines $\frac{x}{3} + \frac{y}{4} + \frac{1}{7} = 0$ and $\frac{2x}{3} + \frac{y}{2} + \frac{1}{10} = 0$ are parallel or perpendicular.
15. Find the slopes of the straight lines $y = 0.7x - 11$ and $x = -11$.
16. **(Compulsory Question)** Find the slope of the straight line perpendicular to the line joining the points $P(a, b)$ and $Q(-a, -b)$.

(Part - C)

(6 × 5 = 30)

Answer any 6 questions. (Question No. 24 is compulsory).

17. Find the area of the quadrilateral formed by the points (8,6), (5,11), (-5,12) and (-4,3).
18. Find the equation of the median drawn from the vertex A of $\triangle ABC$ whose vertices are $A(6, 2)$, $B(-5, -1)$ and $C(1, 9)$.
19. Find the equation of a straight line passing through the point (-3,8) and such that the sum of its positive intercepts on the coordinate axes is 7.
20. Without using Pythagoras theorem, show that the vertices (1, -4), (2, -3) and (4, -7) form a right-angled triangle.
21. Find the equation of a straight line joining the point of intersection of $7x - 3y = -12$ and $2y = x + 3$ to the point of intersection of $3x + y + 2 = 0$ and $x - 2y - 4 = 0$.
22. Find the equation of the straight lines if the sum and product of their intercepts are 1 and -6 respectively.
23. In $\triangle ABC$, the midpoints of sides AB, BC, AC are $P(11, 7)$, $Q(13.5, 4)$, $R(9.5, 4)$ respectively. Find the vertices A, B, C of the triangle.
24. **(Compulsory Question)** If the points $A(a, 0)$, $B(0, b)$ and $C(1, 1)$ are collinear, prove that $\frac{1}{a} + \frac{1}{b} = 1$.

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Comprehensive Master Question Bank

Class: 10

Subject: Mathematics

Exclusive Creative & Application Sums (Not from Textbook)

Unit 1: Relations and Functions

I. Creative Questions (10 Questions)

1. Let $A = \{x \in \mathbb{Z} \mid -2 \leq x < 3\}$ and $B = \{x \in \mathbb{N} \mid x \text{ is a prime factor of } 30\}$. Find $A \times B$ and verify if $n(A \times B) = n(A) \times n(B)$.
2. A relation R on the set of natural numbers $\mathbb{N}$ is defined by $R = \{(x, y) \mid y = 3x - 5, x < 6\}$. Express R as a set of ordered pairs. Determine its domain, co-domain, and range. Is R a function?
3. If the function $f : \mathbb{R} \rightarrow \mathbb{R}$ is defined by $f(x) = \frac{3x-4}{2}$ and $g(x) = \frac{2x+4}{3}$, prove that $(f \circ g)(x) = (g \circ f)(x) = x$.
4. Given $f(x) = x^2 - 4x + 3$ and $g(x) = x + 2$. Find the value of x for which $f(g(x)) = g(f(x))$.
5. Find the domain of the function $f(x) = \frac{\sqrt{x^2-9}}{x-5}$.
6. If $f(x) = 4x + 3$, $g(x) = 2x - 1$, and $h(x) = 3x + 2$, show that $f \circ (g \circ h) = (f \circ g) \circ h$.
7. Let $A = \{1, 2, 3\}$ and $B = \{4, 5, 6, 7\}$. Let $f = \{(1, 5), (2, 6), (3, 4)\}$ be a function from A to B . Identify the type of function and justify your answer.
8. If $f(x) = px + q$ and $f(f(x)) = 9x + 8$, find the possible values of p and q .
9. A function $f : [-5, 7) \rightarrow \mathbb{R}$ is defined as follows:

$$f(x) = \begin{cases} x^2 + 2x; & -5 \leq x < -1 \\ 3x - 4; & -1 \leq x < 3 \\ 2x^2 - 5; & 3 \leq x < 7 \end{cases}$$

Find the value of $\frac{f(-3)+f(2)}{f(4)-f(0)}$.

10. Check whether the function $f : \mathbb{N} \rightarrow \mathbb{N}$ defined by $f(n) = 3n + 2$ is a bijection.

II. Application Questions (5 Questions)

11. **Data Analytics:** A tech company estimates its monthly server maintenance cost using the function $C(x) = 2500 + 45x$, where x is the number of terabytes of data stored. If the company's budget for maintenance is \$5,650, how many terabytes of data can they store?

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12. **Physics (Kinematics):** The velocity of an accelerating car is given by $V(t) = 5t^2 + 2t + 8$ (in m/s), where t is the time in seconds. Find the velocity of the car at $t = 4$ seconds and $t = 6$ seconds.
13. **Economics:** The profit $P(x)$ obtained by selling x units of a smart watch is given by $P(x) = -2x^2 + 400x - 1500$. Find the number of units x that must be sold to achieve a profit of Rs. 18,500.
14. **Cryptography:** A simple encryption function is defined by $E(x) = (7x + 3) \pmod{26}$. If the numerical value of a letter is $x = 12$, find its encrypted value.
15. **Business:** A catering service charges a base fee of Rs. 5000 and Rs. 250 per guest. Write a linear function $F(g)$ for the total cost for g guests. Find the number of guests if the total bill was Rs. 22,500.

Unit 2: Numbers and Sequences

I. Creative Questions (10 Questions)

- Using Euclid's Division Lemma, show that the cube of any positive integer is of the form $7m$, $7m + 1$, or $7m + 6$ for some integer m .
- Find the greatest number consisting of 4 digits which is exactly divisible by 18, 24, and 36.
- Determine the value of x such that $5^{82} \equiv x \pmod{11}$.
- In an Arithmetic Progression (A.P.), the sum of the 5th and 9th terms is 72, and the sum of the 7th and 12th terms is 97. Find the A.P.
- If the p^{th} term of an A.P. is $\frac{1}{q}$ and the q^{th} term is $\frac{1}{p}$, prove that the sum of the first pq terms is $\frac{1}{2}(pq + 1)$.
- Find the sum of all natural numbers between 250 and 800 which are divisible by 7 but not by 3.
- The product of three consecutive terms of a Geometric Progression (G.P.) is 1728 and their sum is 38. Find the terms.
- If a, b, c are in G.P., prove that $\log_x a, \log_x b, \log_x c$ are in A.P.
- Find the sum to n terms of the series: $7 + 77 + 777 + 7777 + \dots$
- Find the sum of the series: $1^2 - 2^2 + 3^2 - 4^2 + \dots + 31^2$.

II. Application Questions (5 Questions)

- Financial Planning:** A man repays a loan of Rs. 1,18,000 by paying Rs. 1000 in the first month and then increases the payment by Rs. 200 every month. How long will it take for him to clear the loan completely?
- Biology (Population Growth):** A bacterial culture doubles its population every 3 hours. If the initial population is 500 bacteria, what will be the population after 24 hours?
- Sports Tournament:** In a massive knockout tennis tournament, 1024 players participate. In each round, half the players are eliminated. Write the sequence of the number of matches played in each round and find the total number of matches played to decide the winner.
- Engineering:** A bouncing ball is dropped from a height of 48 meters. After each bounce, it rebounds to two-thirds of the height from which it fell. Find the total distance travelled by the ball before it comes to rest.
- Architecture:** A pyramid-like structure is built using stone blocks. The top row has 2 blocks, the next has 5 blocks, the next has 8 blocks, and so on. If the structure has 20 rows, find the total number of stone blocks used.

Unit 3: Algebra

I. Creative Questions (10 Questions)

1. Solve the following system of linear equations:

$$\frac{2}{x} + \frac{3}{y} - \frac{1}{z} = 5; \quad \frac{1}{x} - \frac{2}{y} + \frac{3}{z} = 4; \quad \frac{3}{x} + \frac{1}{y} + \frac{2}{z} = 11.$$
2. The LCM and GCD of two polynomials are $(x^3 - 8)(x^2 + 2x + 4)$ and $(x - 2)$ respectively. If one of the polynomials is $x^3 - 8$, find the other polynomial.
3. Simplify: $\frac{x^2-5x+6}{x^2-4} \times \frac{x^2+3x+2}{x^2-2x-3} \div \frac{x^2-4x+4}{x^2-1}$.
4. Find the square root of the polynomial $16x^4 - 24x^3 + 25x^2 - 12x + 4$ using the division method.
5. If the polynomial $mx^4 + nx^3 + 29x^2 - 20x + 4$ is a perfect square, find the values of m and n .
6. Solve the quadratic equation by completing the square method: $3x^2 - 5\sqrt{2}x + 4 = 0$.
7. Find the value of k for which the quadratic equation $(k + 1)x^2 - 2(k - 1)x + 1 = 0$ has real and equal roots.
8. If α and β are the roots of the equation $2x^2 - 5x + 3 = 0$, form a quadratic equation whose roots are $\frac{\alpha}{\beta^2}$ and $\frac{\beta}{\alpha^2}$.
9. Find the value of p if the roots of the equation $x^2 - px + 54 = 0$ are in the ratio 2 : 3.
10. Find the matrix X if $2X + \begin{pmatrix} 3 & -1 \\ 4 & 2 \end{pmatrix} = \begin{pmatrix} 5 & 7 \\ -2 & 8 \end{pmatrix}$.

II. Application Questions (5 Questions)

11. **Aviation:** A flight travels a distance of 1200 km. Due to bad weather, its average speed was reduced by 200 km/h, which increased the flight time by 30 minutes. Find the original planned speed of the flight.
12. **Work & Time:** Pipe A can fill a water tank in 4 hours less time than Pipe B. If both pipes are opened together, the tank is filled in 4.8 hours. Find the time taken by each pipe to fill the tank individually.
13. **Commercial Math:** A merchant bought a certain number of identical items for Rs. 4800. If he had bought 4 more items for the same amount, each item would have cost Rs. 100 less. How many items did he buy initially?
14. **Geometry & Algebra:** The hypotenuse of a right-angled triangle is 2 cm more than twice the shortest side. If the third side is 13 cm less than the hypotenuse, find the lengths of all three sides of the triangle.
15. **Cryptography (Matrices):** Three friends Alice, Bob, and Charlie buy stationary. Alice buys 4 pens, 3 pencils, and 2 erasers for Rs. 75. Bob buys 2 pens, 4 pencils, and 5 erasers for Rs. 80. Charlie buys 3 pens, 2 pencils, and 4 erasers for Rs. 77. Frame this as a matrix equation.

Unit 4 & 6: Geometry and Trigonometry

I. Creative Questions (10 Questions)

1. In a trapezium $PQRS$ with $PQ \parallel RS$, the diagonals intersect at O . If $PO = 3x - 1$, $OR = x + 1$, $QO = 5x - 3$, and $OS = 2x - 1$, find the value of x .
2. In $\triangle ABC$, D is a point on BC such that $\angle ADC = \angle BAC$. Prove that $AC^2 = CB \times CD$.
3. Two chords AB and CD of a circle intersect externally at a point P . If $PA = 12$ cm, $AB = 7$ cm, and $PD = 4$ cm, find the length of CD .
4. Prove that the sum of the squares of the sides of a rhombus is equal to the sum of the squares of its diagonals using Pythagoras Theorem.
5. Prove the identity: $\frac{\cos^3 \theta + \sin^3 \theta}{\cos \theta + \sin \theta} + \frac{\cos^3 \theta - \sin^3 \theta}{\cos \theta - \sin \theta} = 2$.
6. Prove that: $(\sin \alpha \cos \beta + \cos \alpha \sin \beta)^2 + (\cos \alpha \cos \beta - \sin \alpha \sin \beta)^2 = 1$.
7. If $\tan \theta + \sin \theta = m$ and $\tan \theta - \sin \theta = n$, prove that $m^2 - n^2 = 4\sqrt{mn}$.
8. From a point P outside a circle with center O , tangents PA and PB are drawn. Prove that OP is the perpendicular bisector of the chord AB .
9. If $x = a \sec \theta \cos \phi$, $y = b \sec \theta \sin \phi$, and $z = c \tan \theta$, prove that $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$.
10. In $\triangle ABC$, AD is the median and E is the midpoint of AD . If BE is extended, it meets AC at F . Prove that $AF = \frac{1}{3}AC$.

II. Application Questions (5 Questions)

11. **Urban Planning:** Two vertical pillars of heights 15m and 25m stand on a flat ground. If the distance between their feet is 24m, find the length of the wire required to connect their tops tightly.
12. **Navigation:** A ship leaves a port and travels 40 km due North, then changes direction and travels 30 km due East. Using geometry, find the shortest distance between the ship and the port.
13. **Meteorology:** From a point 50 meters above a calm lake, the angle of elevation of a stationary cloud is 30° , and the angle of depression of its reflection in the lake is 60° . Calculate the exact height of the cloud from the surface of the lake.
14. **Aviation Safety:** An observer on the ground spots an airplane flying at a constant altitude of $3000\sqrt{3}$ meters. Initially, the angle of elevation is 60° . After 15 seconds, the angle of elevation changes to 30° . Find the speed of the airplane in km/h.
15. **Surveying:** A surveyor wants to find the width of a river. He stands on one bank and observes a tree exactly opposite on the other bank at an angle of elevation of 45° . He retreats 20 meters horizontally and observes the same tree at an angle of elevation of 30° . Find the width of the river.

Unit 5: Coordinate Geometry

I. Creative Questions (10 Questions)

1. The vertices of a triangle are $A(k, 3)$, $B(2, -4)$, and $C(4, k)$. If the area of the triangle is 14 sq. units, find the possible values of k .
2. Show that the points $(-4, -1)$, $(-2, -4)$, $(4, 0)$, and $(2, 3)$ form a rectangle using slopes and distance formula.
3. Find the equation of the perpendicular bisector of the line segment joining the points $A(-3, 5)$ and $B(7, -1)$.
4. Find the coordinates of the circumcenter of the triangle formed by the vertices $(8, 6)$, $(8, -2)$, and $(2, -2)$.
5. The straight line passing through $(a, 2a)$ and $(-2, 3)$ is parallel to the line $4x + 3y + 5 = 0$. Find the value of a .
6. Find the equation of a line passing through the point of intersection of the lines $3x - 4y + 1 = 0$ and $5x + y - 1 = 0$ and which makes equal intercepts on the coordinate axes.
7. A line intersects the x-axis and y-axis at points P and Q respectively. If the midpoint of PQ is $(4, -5)$, find the equation of the line.
8. The vertices of $\triangle ABC$ are $A(-4, 2)$, $B(2, 6)$, and $C(8, -4)$. Find the equation of the altitude drawn from vertex A to the side BC .
9. Find the coordinates of the foot of the perpendicular drawn from the point $(3, 8)$ to the line $x + 3y - 7 = 0$.
10. Prove algebraically that the straight lines $x - 2y + 5 = 0$, $3x + y - 6 = 0$, and $5x - 3y + 4 = 0$ are concurrent (pass through the same point).

II. Application Questions (5 Questions)

11. **Real Estate:** A plot of land is in the shape of a quadrilateral with corner coordinates at $(-5, -3)$, $(-4, -6)$, $(2, -1)$, and $(1, 2)$ in meters. If the cost of laying grass is Rs. 150 per square meter, calculate the total cost.
12. **Solar Energy:** A solar panel is to be installed along a straight line connecting the points $(2, 5)$ and $(8, 11)$. Sunlight hits the panel perfectly perpendicular at its exact midpoint. Find the straight line equation of the sunlight's path.
13. **Robotics:** A robot moves along a straight path represented by the equation $2x - 3y + 12 = 0$. A charging station is located at the origin $(0, 0)$. Find the shortest distance the robot has to travel to reach the charging station from its path.

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14. **Economics:** A factory's production cost C (in thousands of Rupees) linearly depends on the number of units x produced. If it costs Rs. 8,000 to produce 50 units and Rs. 14,000 to produce 100 units, find the linear equation connecting C and x . Estimate the cost for 150 units.
15. **Civil Engineering:** Two straight roads cross each other forming an intersection. The first road connects points $(1, -2)$ and $(4, 7)$. The second road connects $(-2, 5)$ and $(5, -2)$. Find the exact coordinates of the intersection point where a traffic signal needs to be placed.